Plate IX

## Of Knots

*three closed cords, drawn as the sailor sees them*

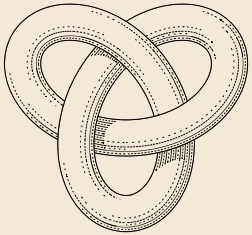


*Fig. 1. The trefoil, of three crossings.*

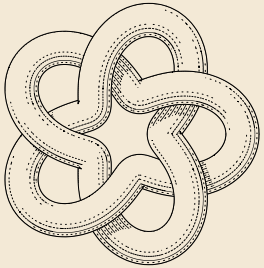


*Fig. 2. The cinquefoil, of five crossings.*

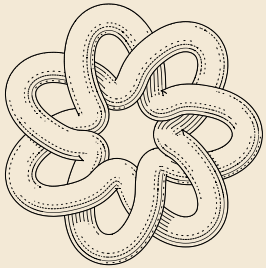


*Fig. 3. The septafoil, of seven crossings.*

These are the first three of the family that a cord makes when wound twice about a ring and turned an odd number of times in going round. Each figure is one formula sampled into a smooth closed path; the library shades the cord as a tube, cuts it wherever it passes beneath itself, and lays the strands one upon another in the order given, so that the eye reads the over and under of the rope at once.

fiziko's own manual calls its knot routine not especially stable. The three cords here are chosen because their crossings are well separated; a knot drawn with crossings close together, or with a strand grazing another, may need its sampling or its width adjusted before the routine finds every crossing.